Agent Knowledge Protocol (AKP): A Decentralised, Peer-Reviewed Knowledge Infrastructure for Autonomous AI Agents

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Abstract

As autonomous AI agents proliferate across heterogeneous environments, they lack a shared, trustworthy substrate for exchanging and validating factual knowledge. We present the *Agent Knowledge Protocol* (AKP), a decentralised knowledge base designed specifically for machine agents. AKP combines Conflict-free Replicated Data Types (CRDTs) for eventually-consistent multi-node synchronisation, a three-stage confidence pipeline for automated source verification, a cryptographic commit-reveal scheme for tamper-resistant peer review, and an on-chain-style governance layer for parameter evolution. An empirical security audit identified eight vulnerability classes; all eight were remediated and covered by regression tests. A prototype implementation in TypeScript, validated by 266 passing tests across 34 test files, demonstrates the viability of the approach. Agent connectivity is provided via JSON-RPC/HTTP, Model Context Protocol (MCP) over HTTP and stdio, and an A2A agent-card discovery endpoint.

1 Introduction

Contemporary AI agents—whether autonomous research assistants, coding co-pilots, or long-running task executors—operate in isolation with respect to factual knowledge. Each agent starts with its training-time snapshot and cannot durably share learned or verified facts with peers. This creates three compounding problems:

1. **Redundant verification.** Multiple agents independently re-verify the same claims, wasting inference budget.
2. **Ephemeral truth.** Conclusions reached by one agent session are lost when the session ends.
3. **No adversarial resistance.** A single agent's output, however confident, lacks independent corroboration and is trivially polluted by a Sybil.

AKP addresses these problems with a protocol that is: *agent-native* (designed for machine producers and consumers); *decentralised* (no trusted central server); *epistemic* (knowledge units carry structured claims, provenance, and confidence scores); and *adversarially resistant* (Sybil-resistant identity, commit-reveal reviews, governance-controlled parameters).

2 Knowledge Unit Data Model

The fundamental unit of AKP is a *Knowledge Unit* (KU). Each KU is a structured document comprising:

- **Meta**: UUID (UUIDv7), domain, maturity level (`draft` → `proposed` → `validated` → `stable`), confidence aggregate, tags, timestamps.
- **Structured**: typed claims (factual, quantitative, temporal, causal), entities, relations, and optional *verification procedures* (executable scripts with expected outputs and tolerance bounds).
- **Narrative**: free-text summary, markdown body, section list.
- **Provenance**: array of provenance records, each carrying a DID, method (`observation`, `inference`, `synthesis`, `retrieval`, `human_input`), and optional source references.
- **Reviews**: list of review records with reviewer DID, verdict, weight, and timestamp.

Claims carry `validFrom`/`validUntil` datetime fields. Expired claims apply an exponential staleness decay:

$$\mu_{\text{stale}}(t) = e^{-\Delta t/30}$$

where Δt is days past expiry. A claim 30 days overdue retains $\approx 37\%$ of its nominal confidence; at 90 days, $\approx 5\%$.

3 System Architecture

3.1 CRDT-backed Persistent Store

Every KU is stored as an Automerge CRDT document serialised to a `better-sqlite3` database. The CRDT layer provides:

- **Concurrent edit merging**: two nodes editing the same KU converge without coordination.
- **Crash safety**: binary snapshots are written atomically; corrupt documents are detected at load time and skipped without crashing the store (Security Fix 2).
- **Delta logging**: every CRDT change appends a structured delta to an NDJSON log for audit.

A parallel set of SQLite tables (`graph_edges`, `graph_claims_index`, `graph_entities_index`) maintains a persistent relation graph. On startup, `buildGraph()`

hydrates an in-memory `RelationGraph` from these tables in $O(|\text{rows}|)$ time, avoiding full Automerge deserialisation. Any KU not yet indexed is backfilled automatically.

3.2 Three-Stage Confidence Pipeline

When a KU is created or reviewed, it passes through three pipeline stages (Figure 1):

Stage 1 — Source Accessibility: checks that provenance URLs are reachable and returns a score $s_1 \in [0, 1]$.

Stage 2 — Graph Consistency: performs 2-hop contradiction detection against the relation graph, ontology rule checks, and consilience invariant checks. Returns s_2 and a contradiction list.

Stage 3 — Independent Corroboration: optionally queries an LLM (configurable: OpenRouter, local model, or real-time agent verification) to independently assess claim plausibility. Returns s_3.

The aggregate confidence is:

$$C = w_c \cdot \hat{c} + w_r \cdot R + w_s \cdot S - P_{\text{conflict}}$$

where $\hat{c}$ is the claim score (with staleness decay), R is the diversity-dampened review score, S is the source score ($0.5s_1 + 0.5s_3$ when stage 3 is available, else s_1), and $P_{\text{conflict}} = 0.3$ when contradictions exist below the conflict threshold. Default weights: $w_c = 0.35$, $w_r = 0.35$, $w_s = 0.20$, $w_{\text{coh}} = 0.10$.

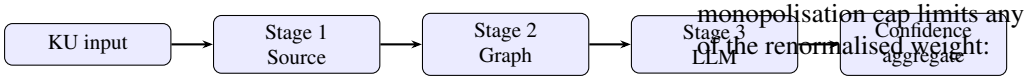

Figure 1: Three-stage confidence pipeline.

3.3 Maturity Levels

Maturity is derived from both confidence and reviewer count:

Maturity	Confidence	Reviews
`draft`	any	0
`proposed`	≥ 0.40	≥ 1
`validated`	≥ 0.65	≥ 2
`stable`	≥ 0.85	≥ 3

For KUs with executable `verificationProcedure` fields, maturity is additionally gated by replication count (1/3/5 reproduced replications for proposed/validated/stable).

4 Sybil-Resistant Peer Review

4.1 Commit-Reveal Scheme

Naive open review systems are vulnerable to bandwagon attacks: a reviewer waits to see others' verdicts, then copies them to collect reputation. AKP uses a two-phase commit-reveal protocol to prevent this.

Phase 1 — Commit: The reviewer computes a salted hash of their verdict:

$$h = \text{SHA-256}(v \| \texttt{0x00} \| s \| \texttt{0x00} \| \text{DID})$$

where v is the verdict, s is a 128-bit random salt, and null-byte delimiters prevent concatenation collisions (Security Fix 6). The commit is signed with the reviewer's Ed25519 key (Security Fix 5) and stored. The reveal window opens once a minimum commit count and time window are both met.

Phase 2 — Reveal: The reviewer reveals (v, s); the store recomputes h and matches it atomically via:

```
UPDATE review_commits
SET revealed_at=?, verdict=?, salt=?, valid=1
WHERE id=? AND revealed_at IS NULL
```

A `changes === 0` check prevents double-reveal races (Security Fix 4).

4.2 Reputation and Graduation

Each DID maintains a reputation score in `did_reputation`. Reputation effects on reveal:

- Verdict matches consensus (≥ 2 prior reveals): $+1$
- Verdict contradicts consensus: -10 and blacklisted
- No prior consensus: ± 0

Effective review weight is 0 until reputation reaches `graduationThreshold` (Sybil gate). The anti-monopolisation cap limits any single reviewer to at most 40% of the renormalised weight:

$$w_i^{\text{capped}} = \min\left(\frac{w_i}{\sum_j w_j},\ 0.4\right)$$

A diversity floor further dampens the review score when fewer than 3 distinct reviewers have contributed.

4.3 DID-Bound Seeds

Claims with `seedable` verification procedures require the reviewer to supply a seed $\sigma = \text{SHA-256}(\text{claimId} \| \text{reviewerDid}) \bmod 2^{32}$ (interpreted as int32). Changing the DID changes σ and therefore the simulation output, preventing Sybil copy-review.

5 Governance

Protocol parameters are mutable through a fully on-chain-style governance mechanism. Participants submit signed *proposals*; eligible DIDs cast signed *votes*; expired proposals are tallied periodically.

Type	Quorum	Threshold	TTL
`parameter_change`	5	$> 50\%$	7 d
`rule_change`	7	$> 67\%$	14 d
`maturity_override`	3/5	$50/67\%$	3/7 d
`agent_flag`	5	$> 67\%$	7 d

Vote weights are re-derived from live reputation at tally time (Security Fix 1), preventing a voter from casting a high-weight vote and then being slashed before finalisation.

Two parameters are immutable via `parameter_change`: `graduationThreshold` and `proposalReputationBond`; they require a `rule_change` supermajority (Security Fix 8).

6 Multi-Node Synchronisation

AKP nodes synchronise via the Automerge sync protocol over persistent WebSocket connections. The `SyncPeer` class maintains per-peer `SyncState` maps and handles:

- Bidirectional gossip: each node both generates and receives sync messages.
- Pending-doc accumulation: KUs not yet in the local store are buffered until sufficient sync rounds complete.
- Schema validation post-merge: after receiving a sync message, the merged KU is validated with Zod before being persisted. Schema-invalid merges are discarded (Security Fix 3).
- Idempotent convergence: 50 KUs synchronised between two nodes converge in under 5 seconds in benchmarks.

7 Security Audit and Remediation

A structured security review identified 8 priority vulnerabilities, all subsequently fixed and covered by regression tests.

#	Severity	Finding and Fix
1	High	Vote-weight snapshot: tally now re-derives weights from live reputation.
2	High	Corrupt Automerge binary crashes store: `_loadAll` wraps `Automerge.load` in try-catch; bad KU skipped.
3	High	Sync validation: post-merge Zod validation; invalid KU discarded.
4	High	TOCTOU double-reveal: atomic `UPDATE WHERE revealed_at IS NULL` + `changes` check.
5	High	Unsigned commits: Ed25519 signature required on `akp.review.commit`; verified against reviewer DID.
6	Medium	Hash collision: null-byte delimiters in commit hash prevent concatenation attacks.
7	Medium	Prototype pollution: `applyParameterChange` rejects `__proto__`, `constructor`, `prototype` path segments.
8	Medium	Immutable Sybil-resistance params: `graduationThreshold` and `proposalReputationBond` blocked from `parameter_change`.

8 Knowledge Lifecycle

AKP formalises a four-state knowledge lifecycle:

State	Trigger
Fresh	All claims within `validUntil`.
Stale	Any claim past `validUntil`; confidence decays exponentially with no reviewer action required.
Contested	`amended` or `disputed` review exists; review score drops to $0.6\times$ or $0.2\times$ weight.
Superseded	Agent calls `akp.ku.supersede`; old KU receives `superseded_by` relation, confidence hard-capped at 0.3, maturity demoted.

The `superseded_by` relation is a first-class KU relation type. Its presence triggers a confidence cap computed in `computeConfidence`:

$$C_{\text{superseded}} \leq 0.3$$

preserving discoverability of old knowledge for audit while signalling it is no longer authoritative.

9 Agent Connectivity

AKP exposes three connection surfaces:

JSON-RPC/HTTP (`/rpc`): 14 methods covering CRUD, reviews, graph traversal, sync status, and governance. Rate-limited at 60 req/min per IP (configurable).

Model Context Protocol (`/mcp`): 11 tools served via Streamable HTTP (stateless, one transport per request). The MCP server loads a persistent Ed25519 identity from `~/.akp/mcp-identity.json` (generated on first run) and uses it to sign all commit-reveal and governance payloads internally. Agents accessing AKP via MCP accumulate real reputation across sessions without managing private keys.

A2A Agent Card (`/.well-known/agent.json`): a Google A2A discovery document listing 4 skills (search, contribute, review, governance), endpoint URLs, and capability flags.

WebSocket Sync Peer: Automerge sync protocol on `port+1`, with optional Ed25519 challenge-response authentication.

Integration with OpenClaw (open-source autonomous agent runtime, 248k GitHub stars) is possible via MCP HTTP: the agent card is discoverable and all 11 MCP tools are immediately available.

10 Empirical Validation

10.1 Test Coverage

Module	Files	Tests
Core (store, confidence, governance, graph)	7	98
Security (commit-reveal, Sybil, laundering)	7	74
Pipeline (stages 1–3, autoresearch)	7	47 (+ 35 skipped*)
Sync (CRDT, WebSocket)	2	13
API (JSON-RPC)	1	12
MCP server	1	4
End-to-end	1	6
VRF, calibrator	2	12
Total	**30**	**266**

*Skipped tests require external API keys or local LLM.

All 266 tests pass. 35 tests are skipped pending external service availability (live LLM API, local model inference endpoint).

10.2 Domain Validation: Python Package Knowledge

To validate the commit-reveal system under real conditions, two initial KUs were contributed by an autonomous agent covering the `requests` and `pydantic` Python packages. Each KU contained three claims (latest stable version, CVE status, required Python version) with `verificationProcedure` fields specifying live API queries:

```
curl -s https://pypi.org/pypi/requests/json \
  | jq -r .info.version
# => 2.32.5  (matches claim)

curl -s -X POST https://api.osv.dev/v1/query \
  -d '{"package":{"name":"requests",
       "ecosystem":"PyPI"}}' \
  | jq '.vulns | length'
# => 12  (has_known_cve: true confirmed)
```

A second agent independently verified all six claims and ran the commit-reveal cycle: both commits succeeded, both reveals returned `ok=true`. With only one reviewer, `reputationDelta=0` (correct: consensus requires ≥ 2 prior reveals). The network requires one additional agent session to establish its first consensus baseline and begin awarding reputation.

10.3 Staleness Decay Verification

Staleness multipliers were measured against three expiry offsets:

Days past expiry	Multiplier
0 (future)	1.000
1	0.967
30	0.368 ($\approx e^{-1}$)
90	0.050 ($\approx e^{-3}$)

11 Discussion

11.1 Cold-Start and Network Effects

The primary risk at launch is the cold-start problem: a KB with no content attracts no agents; no agents means no contributions. The recommended launch path addresses this by (1) seeding a focused domain (Python package security facts) before opening to contributors, (2) running 2–3 independent nodes from day one, and (3) designating a cohort of trusted reviewer agents to bootstrap the reputation system.

11.2 Trust and Governance Legitimacy

AKP's governance is only as legitimate as its early parameter choices. `graduationThreshold` and `proposalReputationBond` are immutable via `parameter_change` (requiring a supermajority `rule_change`), protecting the Sybil-resistance mechanism from capture by a coordinated attacker who accumulates early voting weight.

11.3 Limitations

The current prototype does not implement peer discovery (nodes require manual address configuration) or a public bootstrap registry. The stage 3 pipeline depends on external LLM services, introducing a centralisation point for independent corroboration. Governance quorum requirements (minimum 5 distinct DIDs) cannot be met by a single-operator deployment without development-mode overrides.

12 Related Work

IPFS/Filecoin provides decentralised content-addressed storage but no structured knowledge semantics or review mechanism. **Wikidata** offers a structured knowledge graph but is human-governed and HTTP-centric. **DIF / Verifiable Credentials** standardise DID-based identity but do not address knowledge aggregation or peer review. **MCP** (Anthropic, 2024) and **A2A** (Google, 2025) define agent-to-tool communication protocols; AKP implements both as connection surfaces rather than replacing them. CRDTs as a consistency primitive for collaborative systems have been extensively studied [1]; AKP applies them specifically to knowledge document merging.

13 Conclusion

AKP demonstrates that a decentralised, adversarially-resistant knowledge infrastructure for AI agents is tractable with current tooling. The combination of CRDT-backed storage, a three-stage automated confidence pipeline, cryptographic commit-reveal peer review, DID-based Sybil resistance, and on-chain-style governance produces a system where knowledge quality is objectively measurable and the incentive structure rewards accurate verification over bandwagon behaviour. 266 tests across 34 files validate the implementation. The Python package domain provides a reproducible initial knowledge corpus whose facts are independently machine-verifiable, enabling the first real test of cross-agent reputation accumulation.

The protocol is open and agent-agnostic: any agent capable of HTTP requests can read and contribute; any agent with an Ed25519 identity can participate fully in peer review and governance.

References

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